

OB'YACHEVO, Russian Federation

WMO: 229960

Lat: 60.3287N

Lon: 49.6357E

Elev: 499

StdP: 14.43

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-19.6	-31.2	1.0	-25.8	-24.7	1.4	-19.4	16.5	18.6	15.0	17.3	2.5	10	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	84.2	68.3	80.6	66.1	76.8	64.0	70.1	80.5	68.0	77.5	66.1	74.4	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.3	74.4	64.7	93.2	72.3	62.9	87.4	70.8	34.3	80.2	32.6	77.6	31.1	74.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.0	14.0	12.4	DB	-28.2	89.0	6.5	3.0	-32.9	91.2	-36.6	93.0	-40.3	94.7	-45.0	96.9
(5)				WB	-28.7	73.6	6.7	3.7	-33.5	76.3	-37.4	78.5	-41.1	80.5	-45.9	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	8.8	11.1	23.4	37.3	49.8	59.3	64.3	57.9	48.3	37.1	23.9	14.4
	DBStd	21.58	14.42	13.64	9.32	8.72	9.95	8.35	7.77	7.93	7.32	8.24	10.92	14.59
	HDD50	6162	1278	1089	824	390	132	15	3	17	118	409	784	1103
	HDD65	10588	1743	1509	1289	832	479	210	109	247	502	866	1234	1568
	CDD50	1210	0	0	0	8	125	293	447	263	66	8	0	0
	CDD65	161	0	0	0	0	7	38	88	27	1	0	0	0
	CDH74	1425	0	0	0	2	81	355	772	213	2	0	0	0
	CDH80	358	0	0	0	0	11	76	209	62	0	0	0	0
Wind	WSAvg	6.0	6.1	6.1	6.5	6.3	6.6	5.8	4.9	5.1	5.7	6.3	6.3	6.4
Precipitation	PrecAvg	25.6	1.5	1.2	1.5	1.3	2.2	3.1	2.5	3.3	2.4	2.2	2.0	1.8
	PrecMax	31.1	2.6	2.5	4.6	2.8	5.6	5.5	7.0	6.9	4.5	3.8	4.3	2.8
	PrecMin	20.3	0.7	0.3	0.2	0.1	1.1	1.0	0.4	0.8	0.8	0.9	0.6	1.1
	PrecStd	3.1	0.4	0.6	0.9	0.6	1.1	1.3	1.5	1.5	1.0	0.9	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	35.2	48.0	69.4	81.6	86.9	88.5	87.3	72.1	63.0	46.4	38.5
		MCWB	32.1	32.9	39.1	53.5	62.2	68.0	68.9	69.5	61.5	52.6	44.6	38.0
	2%	DB	32.0	32.8	41.7	60.4	75.4	81.7	85.4	79.2	66.2	55.8	40.4	35.1
		MCWB	31.2	31.0	35.1	47.5	57.5	65.9	68.3	67.2	57.6	50.2	39.0	34.3
	5%	DB	29.7	30.7	38.3	54.7	71.3	78.0	82.1	74.4	62.4	51.6	37.5	33.0
		MCWB	28.8	29.3	33.2	43.7	55.4	63.3	67.8	64.0	55.5	47.4	36.2	32.1
	10%	DB	27.0	28.0	35.4	50.4	66.5	74.2	78.4	70.3	59.3	47.9	35.0	31.3
		MCWB	26.2	26.4	31.8	41.2	53.6	61.3	65.9	61.9	54.3	45.4	34.0	30.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	33.5	39.9	54.3	65.5	71.4	73.3	71.4	62.4	54.2	44.6	38.1
		MCDB	32.9	34.5	46.7	68.5	75.0	82.4	82.3	82.8	69.0	59.5	46.5	38.2
	2%	WB	31.2	31.4	36.5	48.8	60.5	68.3	70.5	68.5	59.6	50.6	39.4	34.6
		MCDB	31.8	32.7	40.1	58.4	70.9	77.9	80.9	77.8	64.6	54.4	40.2	35.1
	5%	WB	28.8	29.3	34.1	45.0	57.5	66.0	68.9	65.6	56.8	48.1	36.5	32.3
		MCDB	29.6	30.8	37.3	52.7	66.2	75.2	78.9	71.8	60.2	51.3	37.3	32.8
	10%	WB	26.3	26.6	32.2	42.2	54.8	63.5	67.2	63.3	54.7	45.4	34.2	30.4
		MCDB	27.2	27.9	34.9	49.1	65.0	71.2	76.1	68.9	58.6	47.7	35.0	31.3
Mean Daily Temperature Range	5% DB	MDBR	10.3	12.1	13.6	15.4	17.9	17.5	17.6	15.3	12.3	7.9	6.8	8.8
		MCDBR	9.1	9.1	15.3	22.8	25.0	22.5	22.7	20.5	18.1	14.2	8.3	5.5
		MCWBR	9.3	8.4	10.5	12.9	12.2	9.9	9.2	9.5	10.3	9.3	7.8	5.8
	5% WB	MCDBR	9.0	8.9	11.6	19.8	21.5	19.6	20.2	17.7	15.1	11.9	7.5	5.5
		MCWBR	9.2	8.4	8.4	12.2	12.2	9.9	9.9	9.3	9.5	8.6	7.5	5.8
Clear-Sky Solar Irradiance	taub	0.264	0.302	0.343	0.375	0.363	0.360	0.393	0.384	0.349	0.324	0.283	0.242	
	taud	2.013	2.053	2.030	2.147	2.415	2.455	2.366	2.372	2.448	2.363	2.120	1.962	
	Ebn at Noon	146	210	240	258	274	276	262	255	246	209	140	103	
	Edh at Noon	15	26	36	39	33	32	34	31	25	19	13	10	
All-Sky Solar Radiation	RadAvg	105	328	706	1176	1553	1646	1666	1179	673	279	96	54	
	RadStd	16	81	113	156	159	145	172	128	92	36	18	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page